#1

```
import pandas as pd


# Load the dataset
df = pd.read_csv("/content/WA_Fn-UseC_-HR-Employee-Attrition-selected-columns.csv")


# Number of rows and columns
print("----- Dataset Shape -----")
print("Number of Rows    :", df.shape[0])
print("Number of Columns :", df.shape[1])


# Data types
print("\n----- Data Types -----")
print(df.dtypes)


# Missing values
print("\n----- Missing Values -----")
print(df.isnull().sum())


# Statistical summary
print("\n----- Statistical Summary -----")
print(df.describe(include='all'))
```

```
----- Dataset Shape -----
Number of Rows    : 124
Number of Columns : 10

----- Data Types -----
```

OUTPUT:

```
Age                 float64
Attrition           float64
BusinessTravel      float64
DailyRate           float64
Department          float64
DistanceFromHome    float64
Education           float64
EducationField      float64
EmployeeCount       float64
EmployeeNumber      float64
dtype: object

----- Missing Values -----
Age                 124
Attrition           124
BusinessTravel      124
DailyRate           124
Department          124
DistanceFromHome    124
Education           124
EducationField      124
EmployeeCount       124
EmployeeNumber      124
dtype: int64

----- Statistical Summary -----
       Age  Attrition  BusinessTravel  DailyRate  Department  \
count  0.0        0.0             0.0        0.0         0.0
mean   NaN        NaN             NaN        NaN         NaN
std    NaN        NaN             NaN        NaN         NaN
min    NaN        NaN             NaN        NaN         NaN
25%    NaN        NaN             NaN        NaN         NaN
50%    NaN        NaN             NaN        NaN         NaN
75%    NaN        NaN             NaN        NaN         NaN
max    NaN        NaN             NaN        NaN         NaN

       DistanceFromHome  Education  EducationField  EmployeeCount  \
count               0.0        0.0             0.0            0.0
mean                NaN        NaN             NaN            NaN
std                 NaN        NaN             NaN            NaN
min                 NaN        NaN             NaN            NaN
25%                 NaN        NaN             NaN            NaN
50%                 NaN        NaN             NaN            NaN
75%                 NaN        NaN             NaN            NaN
```

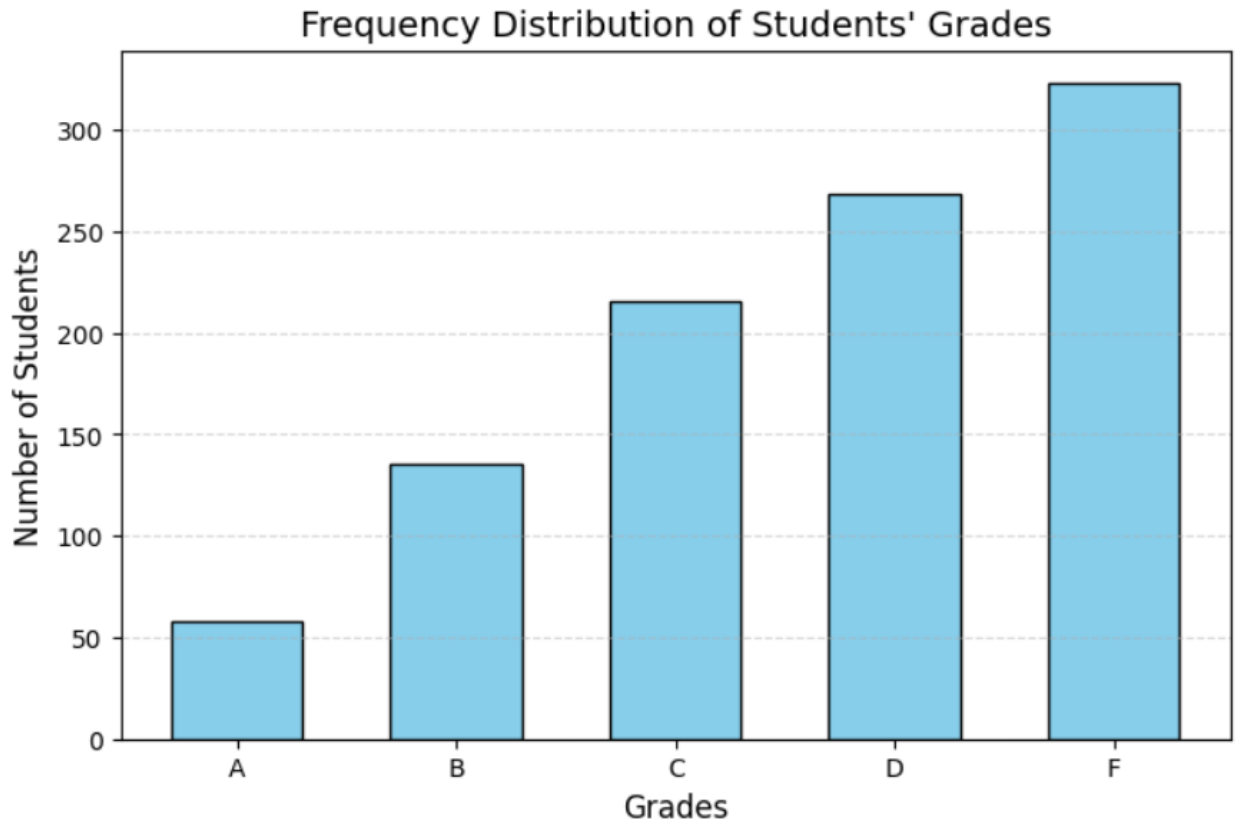
```
max              NaN       NaN           NaN           NaN

      EmployeeNumber
count             0.0
mean              NaN
std               NaN
min               NaN
25%               NaN
50%               NaN
75%               NaN
max               NaN
```

```
#2 Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read Dataset
df = pd.read_csv("/content/StudentsPerformance.csv")

# Display first 5 rows
print("First 5 Records:")
print(df.head())

# Display column names
print("\nColumn Names:")
print(df.columns)

# Create Grade Column
def get_grade(score):
    if score >= 90:
        return 'A'
    elif score >= 80:
        return 'B'
    elif score >= 70:
        return 'C'
    elif score >= 60:
        return 'D'
    else:
        return 'F'

df['Grade'] = df['math score'].apply(get_grade)

# Frequency Distribution
```

```
grade_count = df['Grade'].value_counts().sort_index()

print("\nFrequency Distribution of Grades:")
print(grade_count)

# Plot Bar Chart
plt.figure(figsize=(8,5))

plt.bar(
    grade_count.index,
    grade_count.values,
    color='skyblue',
    edgecolor='black',
    width=0.6
)

plt.title("Frequency Distribution of Students' Grades", fontsize=14)
plt.xlabel("Grades", fontsize=12)
plt.ylabel("Number of Students", fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.5)

plt.show()
```

```
First 5 Records:
   gender race/ethnicity parental level of education         lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B            master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test preparation course  math score  reading score  writing score
0                    none         72             72             74
1               completed         69             90             88
2                    none         90             95             93
3                    none         47             57             44
4                    none         76             78             75

Column Names:
Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
       'test preparation course', 'math score', 'reading score',
       'writing score'],
      dtype='object')

Frequency Distribution of Grades:
Grade
```

A 58
B 135
C 216
D 268
F 323
Name: count, dtype: int64



#3 Upload Dataset

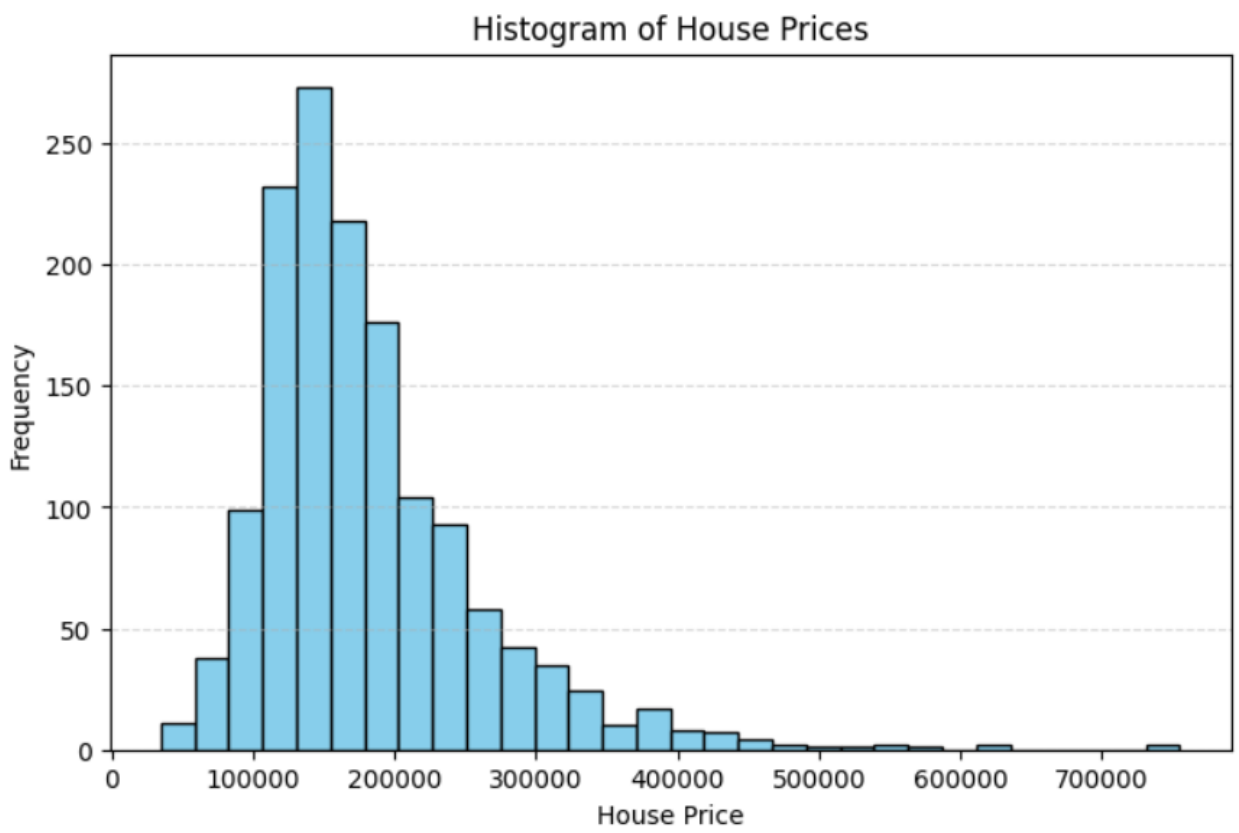
```
from google.colab import files
uploaded = files.upload()

# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read Dataset
df = pd.read_csv("/content/train.csv")

# Display first 5 rows
print("First 5 Records:")
print(df.head())
```

```
# Display column names
print("\nColumns:")
print(df.columns)

# Histogram
plt.figure(figsize=(8,5))
plt.hist(df["SalePrice"],
        bins=30,
        color="skyblue",
        edgecolor="black")

plt.title("Histogram of House Prices")
plt.xlabel("House Price")
plt.ylabel("Frequency")
plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()

# Check skewness
skewness = df["SalePrice"].skew()

print("\nSkewness:", round(skewness,2))

if skewness > 0:
   print("Conclusion: The data is Positively Skewed (Right Skewed).")
elif skewness < 0:
   print("Conclusion: The data is Negatively Skewed (Left Skewed).")
else:
   print("Conclusion: The data is Normally Distributed.")
```

OUTPUT:

```
First 5 Records:
   Id  MSSubClass MSZoning  LotFrontage  LotArea Street Alley LotShape  \
0   1          60       RL         65.0     8450   Pave   NaN      Reg
1   2          20       RL         80.0     9600   Pave   NaN      Reg
2   3          60       RL         68.0    11250   Pave   NaN      IR1
3   4          70       RL         60.0     9550   Pave   NaN      IR1
4   5          60       RL         84.0    14260   Pave   NaN      IR1

  LandContour Utilities  ... PoolArea PoolQC Fence MiscFeature MiscVal MoSold  \
0         Lvl    AllPub  ...        0    NaN   NaN         NaN       0      2
1         Lvl    AllPub  ...        0    NaN   NaN         NaN       0      5
2         Lvl    AllPub  ...        0    NaN   NaN         NaN       0      9
3         Lvl    AllPub  ...        0    NaN   NaN         NaN       0      2
```

```
4       Lvl    AllPub  ...        0    NaN   NaN        NaN      0     12

  YrSold  SaleType  SaleCondition  SalePrice
0   2008        WD         Normal     208500
1   2007        WD         Normal     181500
2   2008        WD         Normal     223500
3   2006        WD        Abnorml     140000
4   2008        WD         Normal     250000

[5 rows x 81 columns]

Columns:
Index(['Id', 'MSSubClass', 'MSZoning', 'LotFrontage', 'LotArea', 'Street',
       'Alley', 'LotShape', 'LandContour', 'Utilities', 'LotConfig',
       'LandSlope', 'Neighborhood', 'Condition1', 'Condition2', 'BldgType',
       'HouseStyle', 'OverallQual', 'OverallCond', 'YearBuilt', 'YearRemodAdd',
       'RoofStyle', 'RoofMatl', 'Exterior1st', 'Exterior2nd', 'MasVnrType',
       'MasVnrArea', 'ExterQual', 'ExterCond', 'Foundation', 'BsmtQual',
       'BsmtCond', 'BsmtExposure', 'BsmtFinType1', 'BsmtFinSF1',
       'BsmtFinType2', 'BsmtFinSF2', 'BsmtUnfSF', 'TotalBsmtSF', 'Heating',
       'HeatingQC', 'CentralAir', 'Electrical', '1stFlrSF', '2ndFlrSF',
       'LowQualFinSF', 'GrLivArea', 'BsmtFullBath', 'BsmtHalfBath', 'FullBath',
       'HalfBath', 'BedroomAbvGr', 'KitchenAbvGr', 'KitchenQual',
       'TotRmsAbvGrd', 'Functional', 'Fireplaces', 'FireplaceQu', 'GarageType',
       'GarageYrBlt', 'GarageFinish', 'GarageCars', 'GarageArea', 'GarageQual',
       'GarageCond', 'PavedDrive', 'WoodDeckSF', 'OpenPorchSF',
       'EnclosedPorch', '3SsnPorch', 'ScreenPorch', 'PoolArea', 'PoolQC',
       'Fence', 'MiscFeature', 'MiscVal', 'MoSold', 'YrSold', 'SaleType',
       'SaleCondition', 'SalePrice'],
      dtype='object')

Skewness: 1.88
Conclusion: The data is Positively Skewed (Right Skewed).
```



Skewness: 1.88
Conclusion: The data is Positively Skewed (Right Skewed).

#4 Upload Dataset

```
from google.colab import files
uploaded = files.upload()

# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read Dataset
df = pd.read_csv("/content/Salary_dataset.csv")

# Display first 5 records
print(df.head())

# Display column names
print(df.columns)

# Box Plot
```

```
plt.figure(figsize=(7,5))
plt.boxplot(df["Salary"], patch_artist=True)

plt.title("Box Plot of Salary Data")
plt.ylabel("Salary")
plt.grid(True)

plt.show()

# Detect Outliers using IQR
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Salary"] < lower) | (df["Salary"] > upper)]

print("\nLower Limit:", lower)
print("Upper Limit:", upper)
print("\nNumber of Outliers:", len(outliers))
print("\nOutliers:")
print(outliers)



Saving Salary_dataset.csv to Salary_dataset (1).csv
   Unnamed: 0  YearsExperience   Salary
0           0              1.2  39344.0
1           1              1.4  46206.0
2           2              1.6  37732.0
3           3              2.1  43526.0
4           4              2.3  39892.0
Index(['Unnamed: 0', 'YearsExperience', 'Salary'], dtype='object')
```

Lower Limit: -9014.25
Upper Limit: 166281.75

Number of Outliers: 0

Outliers:
Empty DataFrame
Columns: [Unnamed: 0, YearsExperience, Salary]
Index: []

#5 Upload Dataset

```
from google.colab import files
uploaded = files.upload()

# Import Libraries
import pandas as pd

# Read Dataset
df = pd.read_csv("StudentsPerformance.csv")

# Display first 5 records
print(df.head())

# Display column names
```

```
print("\nColumns:")
print(df.columns)

# Select Marks Column (Math Score)
marks = df["math score"]

# Calculate IQR
Q1 = marks.quantile(0.25)
Q3 = marks.quantile(0.75)
IQR = Q3 - Q1

# Calculate Lower and Upper Limits
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("\nLower Limit:", lower)
print("Upper Limit:", upper)

# Detect Outliers
outliers = df[(marks < lower) | (marks > upper)]

print("\nNumber of Outliers:", len(outliers))

print("\nOutliers:")
print(outliers[["math score"]])
```

OUTPUT:

```
Saving StudentsPerformance (1).csv to StudentsPerformance (1) (1).csv
   gender race/ethnicity parental level of education         lunch  \
0  female        group B          bachelor's degree      standard
1  female        group C               some college      standard
2  female        group B            master's degree      standard
3    male        group A         associate's degree  free/reduced
4    male        group C               some college      standard

  test preparation course  math score  reading score  writing score
0                    none         72             72             74
1               completed         69             90             88
2                    none         90             95             93
3                    none         47             57             44
4                    none         76             78             75

Columns:
```

```
Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
       'test preparation course', 'math score', 'reading score',
       'writing score'],
      dtype='object')

Lower Limit: 27.0
Upper Limit: 107.0

Number of Outliers: 8

Outliers:
     math score
17           18
59            0
145          22
338          24
466          26
787          19
842          23
980           8

#6 Upload Dataset
from google.colab import files
uploaded = files.upload()

# Import Libraries
import pandas as pd

# Read Dataset
df = pd.read_csv("/content/Test-selected-columns.csv")

# Display first 5 rows
print("First 5 Records:")
print(df.head())

# Check column names
print("\nColumns in Dataset:")
print(df.columns)

# Select Sales column
sales = df["Item_MRP"]

# Calculate Q1, Q3 and IQR
Q1 = sales.quantile(0.25)
Q3 = sales.quantile(0.75)
IQR = Q3 - Q1
```

```
# Calculate lower and upper limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("\nLower Limit:", lower_limit)
print("Upper Limit:", upper_limit)

# Detect outliers
outliers = df[(sales < lower_limit) | (sales > upper_limit)]

print("\nNumber of Outliers:", len(outliers))

# Remove outliers
cleaned_df = df[(sales >= lower_limit) & (sales <= upper_limit)]

print("\nOriginal Dataset Shape:", df.shape)
print("Cleaned Dataset Shape:", cleaned_df.shape)

print("\nFirst 10 Rows of Cleaned Dataset:")
print(cleaned_df.head(10))
```

OUTPUT:

First 5 Records:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	\
0	FDW58	20.750	Low Fat	0.007565	Snack Foods	
1	FDW14	8.300	reg	0.038428	Dairy	
2	NCN55	14.600	Low Fat	0.099575	Others	
3	FDQ58	7.315	Low Fat	0.015388	Snack Foods	
4	FDY38	NaN	Regular	0.118599	Dairy	

	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	\
0	107.8622	OUT049	1999	Medium	
1	87.3198	OUT017	2007	NaN	
2	241.7538	OUT010	1998	NaN	
3	155.0340	OUT017	2007	NaN	
4	234.2300	OUT027	1985	Medium	

	Outlet_Location_Type
0	Tier 1
1	Tier 2
2	Tier 3
3	Tier 2
4	Tier 3

Columns in Dataset:
Index(['Item_Identifier', 'Item_Weight', 'Item_Fat_Content', 'Item_Visibility',
'Item_Type', 'Item_MRP', 'Outlet_Identifier',
'Outlet_Establishment_Year', 'Outlet_Size', 'Outlet_Location_Type'],
dtype='object')

Lower Limit: -43.00989999999999
Upper Limit: 323.44849999999997

Number of Outliers: 0

Original Dataset Shape: (5681, 10)
Cleaned Dataset Shape: (5681, 10)

First 10 Rows of Cleaned Dataset:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	\
0	FDW58	20.750	Low Fat	0.007565	
1	FDW14	8.300	reg	0.038428	
2	NCN55	14.600	Low Fat	0.099575	
3	FDQ58	7.315	Low Fat	0.015388	
4	FDY38	NaN	Regular	0.118599	
5	FDH56	9.800	Regular	0.063817	
6	FDL48	19.350	Regular	0.082602	
7	FDC48	NaN	Low Fat	0.015782	
8	FDN33	6.305	Regular	0.123365	
9	FDA36	5.985	Low Fat	0.005698	

	Item_Type	Item_MRP	Outlet_Identifier	\
0	Snack Foods	107.8622	OUT049	
1	Dairy	87.3198	OUT017	
2	Others	241.7538	OUT010	
3	Snack Foods	155.0340	OUT017	
4	Dairy	234.2300	OUT027	
5	Fruits and Vegetables	117.1492	OUT046	
6	Baking Goods	50.1034	OUT018	
7	Baking Goods	81.0592	OUT027	
8	Snack Foods	95.7436	OUT045	
9	Baking Goods	186.8924	OUT017	

	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type
0	1999	Medium	Tier 1
1	2007	NaN	Tier 2
2	1998	NaN	Tier 3
3	2007	NaN	Tier 2
4	1985	Medium	Tier 3
5	1997	Small	Tier 1

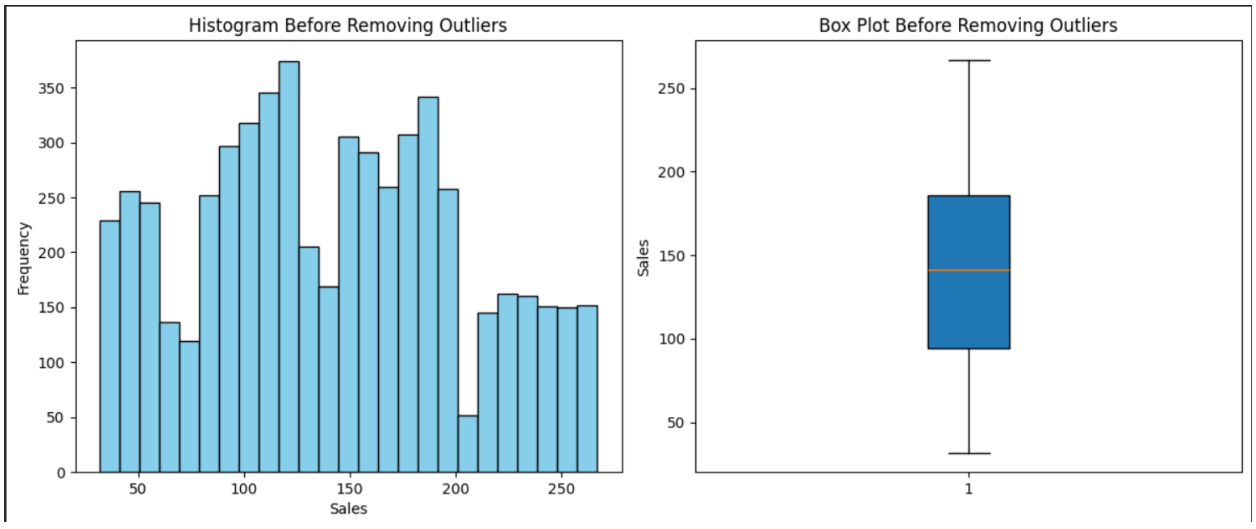
```
6                2009     Medium          Tier 3
7                1985     Medium          Tier 3
8                2002        NaN          Tier 2
9                2007        NaN          Tier 2
```

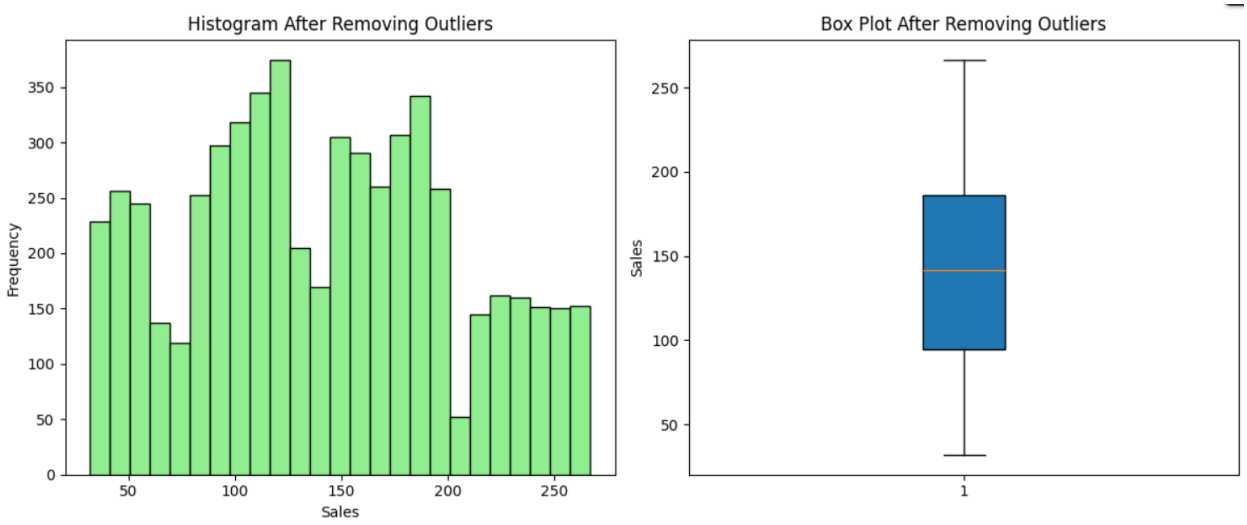
#7 Upload Dataset

```
from google.colab import files
uploaded = files.upload()

# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read Dataset
df = pd.read_csv("/content/Test-selected-columns.csv")

# Display first 5 records
print("First 5 Records:")
print(df.head())

# Select Sales Column
sales = df["Item_MRP"]

# Calculate IQR
Q1 = sales.quantile(0.25)
Q3 = sales.quantile(0.75)
IQR = Q3 - Q1

# Lower and Upper Limits
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

# Remove Outliers
cleaned_df = df[(sales >= lower) & (sales <= upper)]

print("\nOriginal Dataset Shape:", df.shape)
print("Cleaned Dataset Shape:", cleaned_df.shape)

# Create Plots
plt.figure(figsize=(12,10))

# Histogram Before
plt.subplot(2,2,1)
plt.hist(df["Item_MRP"], bins=25,
```

```
        color="skyblue", edgecolor="black")
plt.title("Histogram Before Removing Outliers")
plt.xlabel("Sales")
plt.ylabel("Frequency")

# Box Plot Before
plt.subplot(2,2,2)
plt.boxplot(df["Item_MRP"], patch_artist=True)
plt.title("Box Plot Before Removing Outliers")
plt.ylabel("Sales")

# Histogram After
plt.subplot(2,2,3)
plt.hist(cleaned_df["Item_MRP"], bins=25,
        color="lightgreen", edgecolor="black")
plt.title("Histogram After Removing Outliers")
plt.xlabel("Sales")
plt.ylabel("Frequency")

# Box Plot After
plt.subplot(2,2,4)
plt.boxplot(cleaned_df["Item_MRP"], patch_artist=True)
plt.title("Box Plot After Removing Outliers")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()
```

OUTPUT:

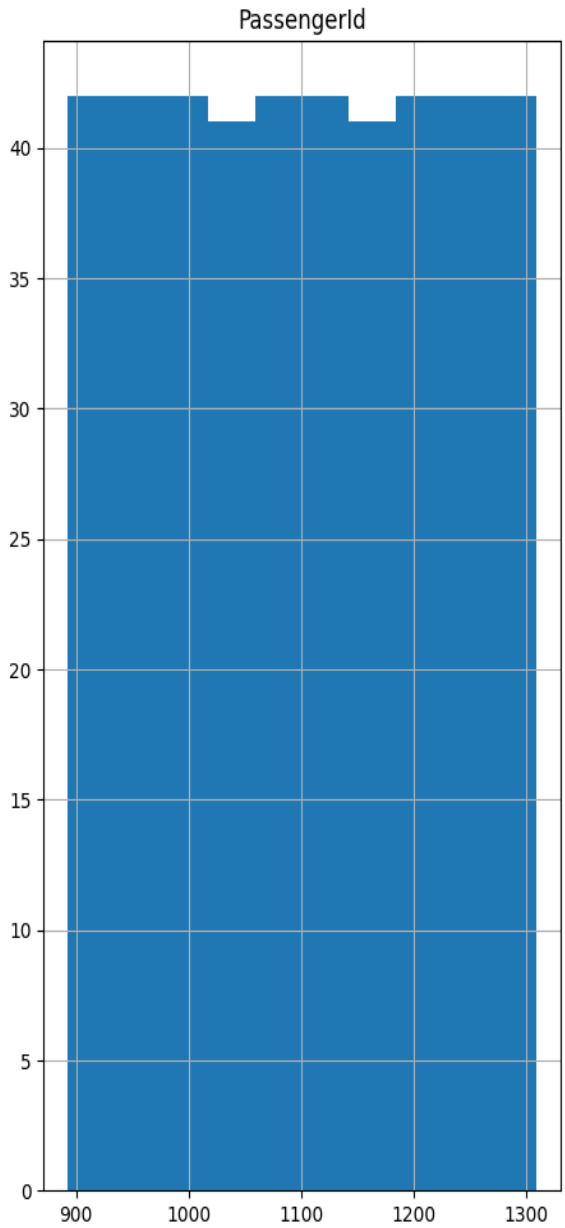


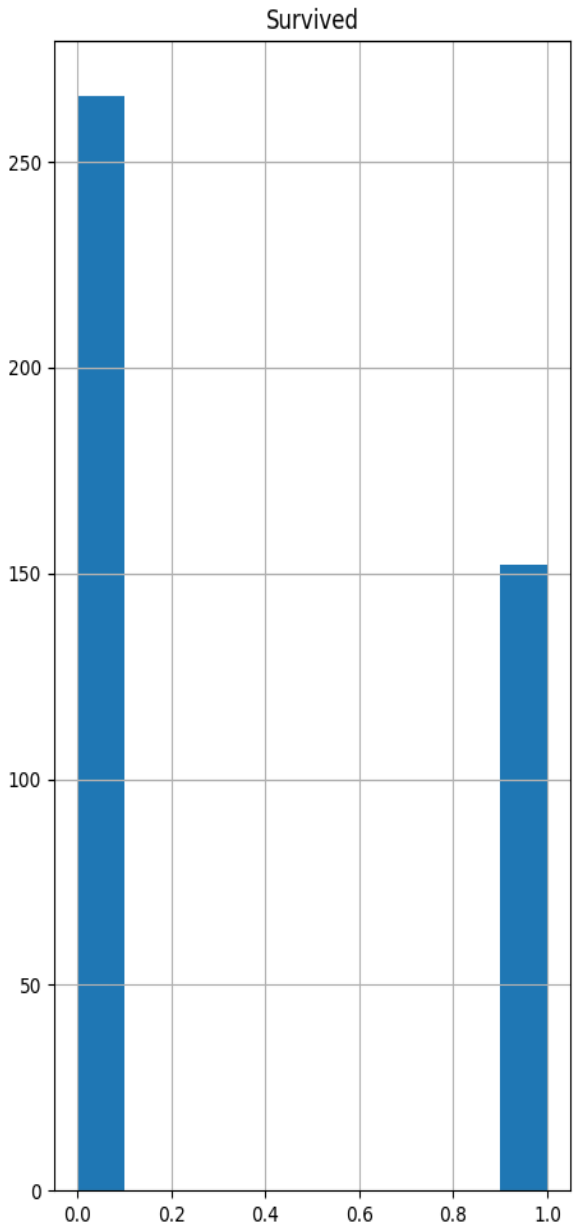


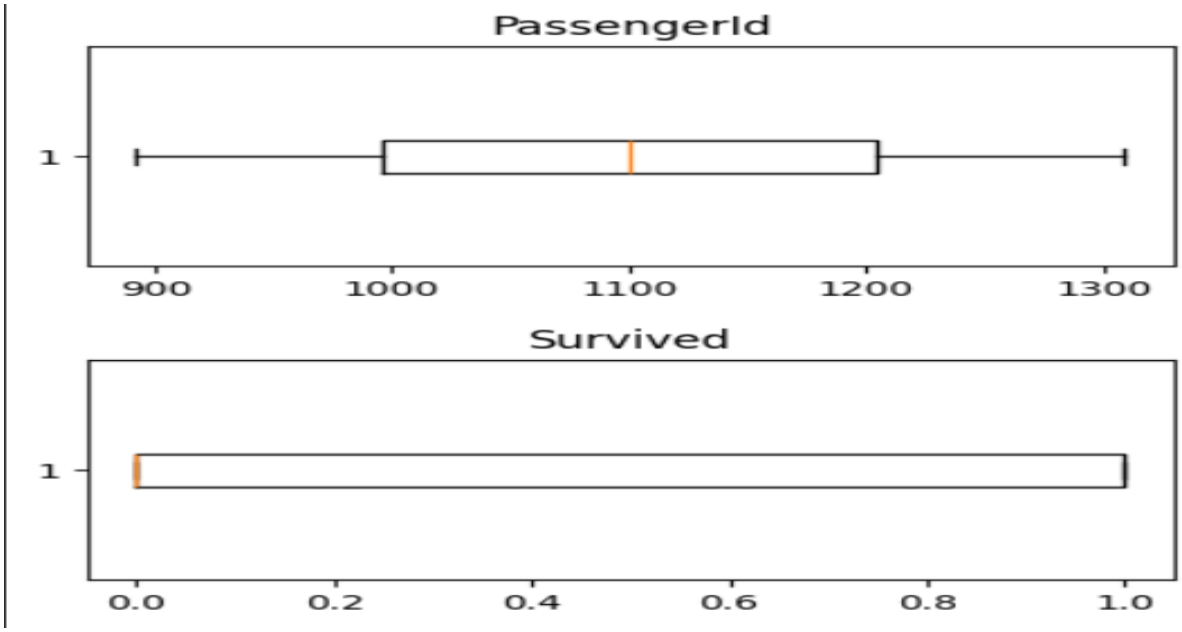
#8

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("/content/gender_submission.csv")

# Dataset Information
print(df.info())

# Missing Values
print(df.isnull().sum())

# Descriptive Statistics
print(df.describe())

# Histograms
df.hist(figsize=(12,10))
plt.show()

# Box Plots
num_cols = df.select_dtypes(include="number").columns
for col in num_cols:
    plt.figure(figsize=(5,2))
    plt.boxplot(df[col].dropna(), vert=False)
    plt.title(col)
    plt.show()

# Detect Outliers
for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
```

```
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]
    print(col, ":", len(outliers))

# Remove Outliers
clean_df = df.copy()
for col in num_cols:
    Q1 = clean_df[col].quantile(0.25)
    Q3 = clean_df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    clean_df = clean_df[(clean_df[col] >= lower) & (clean_df[col] <= upper)]

# Save Cleaned Dataset
clean_df.to_csv("Titanic_Cleaned.csv", index=False)

print(clean_df.shape)


<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 2 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  418 non-null    int64
 1   Survived     418 non-null    int64
dtypes: int64(2)
memory usage: 6.7 KB
None
PassengerId    0
Survived       0
dtype: int64
       PassengerId    Survived
count   418.000000  418.000000
mean   1100.500000    0.363636
std     120.810458    0.481622
min     892.000000    0.000000
25%     996.250000    0.000000
50%    1100.500000    0.000000
75%    1204.750000    1.000000
max    1309.000000    1.000000
```


PassengerId
40
35
30
25
20
15
10
5
0
900
1000
1100
1200
1300


Survived
250
200
150
100
50
0
0.0
0.2
0.4
0.6
0.8
1.0



PassengerId : 0
Survived : 0
(418, 2)